

[F] hsize=T + parshape 全窄 (5 项) ——应与 [A]/[C]

断行一致

已知

正方体 $ABCD-A_1B_1C_1D_1$ 中,

$$\overrightarrow{A_1E} = \frac{1}{4} \overrightarrow{A_1C_1},$$

若 $\overrightarrow{AE} = x\overrightarrow{AA_1} + y(\overrightarrow{AB} + \overrightarrow{AD})$, 则

$x = \underline{\hspace{2cm}}$, $y = \underline{\hspace{2cm}}$.

[G] hsize=T + parshape 窄 4+ 通栏 (长文本, 看图底
以下行是否真通栏)

[详解] 过点 B, D 分别向 AC

作垂线, 垂足分别为 M, N ,

则可得 $AM = \frac{1}{2}$, $BM = \frac{\sqrt{3}}{2}$,

$CN = \frac{1}{2}$, $DN = \frac{\sqrt{3}}{2}$,

$MN = 1$. 由于 $\overrightarrow{BD} = \overrightarrow{BM} + \overrightarrow{MN} + \overrightarrow{ND}$, 所以

$$\begin{aligned} |\overrightarrow{BD}|^2 &= (\overrightarrow{BM} + \overrightarrow{MN} + \overrightarrow{ND})^2 \\ &= |\overrightarrow{BM}|^2 + |\overrightarrow{MN}|^2 + |\overrightarrow{ND}|^2 + 2(\overrightarrow{BM} \cdot \overrightarrow{MN} + \overrightarrow{MN} \cdot \overrightarrow{ND} + \overrightarrow{BM} \cdot \overrightarrow{ND}) \\ &= \left(\frac{\sqrt{3}}{2}\right)^2 + 1^2 + \left(\frac{\sqrt{3}}{2}\right)^2 + 2 \times (0 + 0 + 0) = \frac{5}{2}, \text{ 所以} \\ |\overrightarrow{BD}| &= \frac{\sqrt{10}}{2}. \text{ 故选: } A. \end{aligned}$$

[H] 对照: 无 hsize (parshape 窄 4+ 通栏, 长文本)

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